

PERSONAL INFORMATION

FRANCESCO DONNARUMMA



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Profile	ID	H-index	CIT	DOCS
Google Scholar	cz9mq4AAAAJ	24	2'886	74
Scopus	36665940900	16	925	49
Web of Science	F-6894-2014	16	795	45
OpenAlex	A5048689229	20	1'915	88

Date of birth 12 May 1977 | Nationality Italian

EDUCATION

2010 Ph.D. in Computer and Information Science

February 17th

Università degli Studi di Napoli Federico II

Thesis Title: A model for Programmability and Virtuality in Dynamical Neural Networks

2006 M.Sc. in Physics (Cybernetics),

October 25th

Università degli Studi di Napoli Federico II

Thesis Title: Studio di un controllo senso-motorio ispirato al biologico realizzato tramite reti CTRNN

CURRENT POSITIONS

2023-present First Researcher (Senior Researcher - permanent position)

Institute of Cognitive Sciences and Technologies (ISTC)

National Research Council (CNR) of Italy

via Gian Domenico Romagnosi n. 18/A, 00196 Rome, Italy

PREVIOUS POSITIONS

2018-2022 Researcher (permanent position)

Institute of Cognitive Sciences and Technologies (ISTC)

National Research Council (CNR) of Italy

via Gian Domenico Romagnosi n. 18/A, 00196 Rome, Italy

2016-2018 Researcher (fixed term)

Institute of Cognitive Sciences and Technologies (ISTC)

National Research Council (CNR) of Italy

Contract - in "Probabilistic and Dynamical Systems neuro-computational models for goal-directed behavior of artificial agents and biological organisms" for:

- **Human Science Frontiers Project (HSFP)** grant agreement RGY0088/2014 - Beyond simple choices: computational and neuronal mechanisms for complex spatial behaviors (August 2016 - July 2018)
- **HiPlan: Hierarchical Planning during spatial navigation** funded within - **Human Brain Project (HBP)** second grant agreement (SGA2) - Horizon 2020 Framework Programme for Research and Innovation - Grant number: No. 78590 (August 2018 - December 2018)

2018 Visiting Researcher

Centre d'Etudes des Transformations des Activités Physiques et Sportives (**CETAPS**) Lab
 Université de Rouen Normandie, France, collaborating with the group of Ludovic Seifert -
 (March-April)

2011-2016 Research Fellow

Institute of Cognitive Sciences and Technologies (ISTC)

National Research Council (CNR) of Italy

Contract - in "Dynamical Systems Computational models of goal-directed behaviors and their neural bases" (November 15 - July 31) for:

- **Goal-Leaders** - Goal-directed, Adaptive Builder Robots - Eu project FP7 - ICT - 270108 - (2011-2014)
- **HUMANOBS** - Humanoids That Learn Socio-Communicative Skills Through Observation - Eu project FP7 - ICT - 231453 - (2011-2012)

2010-2011 Research Collaborator

Contract: two-years Post Doc Grant (May 2010 - November 2011) in "Robotics - Environment Reconstruction and Representation for Aerial Vehicles" for:

- **AIRobots** (Innovative aerial service robots for remote inspections by contact) - Eu project FP7 - ICT - 248669 - at the Dipartimento di Informatica e Sistemistica (DIS), Università degli Studi di Napoli Federico II

2009-2010 Research Collaborator

Contract (September 2009 - April 2010) in "Neural Networks and Dynamical Systems in the modelling of Biological Neuronal Networks" for:

- "Modelli computazionali e robotici del comportamento biologico adattativo: problemi metodologici e integrazione multidisciplinare" - **PRIN07**
 Dipartimento di Scienze Fisiche, Università degli Studi di Napoli Federico II.

2009 Research Collaborator

Contract (October - December) in "Projecting and prototyping Evolutionary adaptive filters for electrical lift-controlling" for:

- **EVODIALIFT** (Evolutionary e-diagnostics and Tele-Monitoring for Lift Predictive e-maintenance) Regione Campania (POR 3.17 ICT)
 Dipartimento di Ingegneria dei Materiali (DIMPS), Università degli Studi di Napoli Federico II

2009 Visiting Ph.D.

École Normale Supérieure De Cachan in Paris (July - August), funded by:

- short visit Grant n. 3014 with **GAMES** Eu research networking programme with the group of Patricia Bouyer of the Information Science Laboratory "Laboratoire Spécification et Vérification" (LSV)

2008-2009 Research Collaborator

Contract (November 2008 - August 2009) in "Simulation Techniques for on-line optimization of Event Systems" for:

- **CAWSYS**, project - Bando per la concessione degli aiuti alle PMI in attuazione della misura 3.17 del POR Campania 2000/2006 and eu-cofunded (Decreto Dirigenziale n. 25 del 28/02/2008, Regione Campania, capofila società Nettuno Solutions s.r.l.) at the Dipartimento di Informatica e Sistemistica (DIS), Università degli Studi di Napoli Federico II.

2008 Research Collaborator

Contract (February - April) in "Personal Robotics and Brain-Computer Interfaces. State of the art, costs, benefits and legal issues" for:

- **ETHICBOTS** - Emerging Technoethics of Human Interaction with Communication, Bionic, and robOTic systems - Eu project FP6 - SOCIETY - 17759 - (2005-2008) coordinated by the Dipartimento di Scienze Fisiche, Università degli Studi di Napoli Federico II.

2007-2008 Research Collaborator

Contract (October - January) in "Machine Learning and AI and Robotic Systems: Epistemic Risk Evaluation and Responsibility Problems" for:

- **ETHICBOTS** - Emerging Technoethics of Human Interaction with Communication, Bionic, and robOTic systems - Eu project FP6 - SOCIETY - 17759 - (2005-2008) coordinated by the Dipartimento di Scienze Fisiche, Università degli Studi di Napoli Federico II.

FELLOWSHIPS AND AWARDS**2022-present ISTC Principal Investigator**

Project title: **Free energy principle and the brain: neuronal and phylogenetic mechanisms of Bayesian inference**

Funding Agency: Ministero dell'Università e della ricerca - MUR

Call: PRIN2020.0003357.26-01-2021 PROGETTI DI RICERCA DI RILEVANTE INTERESSE NAZIONALE – Bando 2020 - **PRIN2020**

Expected duration: 36 months

Unit: ISTC-CNR

Total Funding: 709'308

Unit Funding: Euro 169'621

Grant Agreement: 2020529PCP

starting: 01/06/2022

2024-2025 Co-Principal Investigator

Project title: **ACT-SENS: An active inference model of sensorimotor functions and dysfunctions in the brain**

Funding agency: MUR, PNRR M4C2 I1.3 Partenariato Esteso MNESYS- Bando a Cascata SPOKE 4 - "A multi-scale integrated approach to the study of the nervous system in health and disease"

Funding: Euro 300'000

Grant Agreement: No. J33C22002970002

2024-2036 Full Professor Habilitation

role: **Full Professor**

Institution: Italian National Scientific Habilitation

Academic Recruitment Field: 11/E1 - General Psychology, Psychobiology and Psychometrics (current equivalent: GSD 11/PSIC-01)

period: from 06/11/2024 to 06/11/2036

2019-2030 Associate Professor Habilitation

role: **Associate Professor**

Institution: Italian National Scientific Habilitation

Academic Recruitment Field: 11/E1 - General Psychology, Psychobiology and Psychometrics (current equivalent: GSD 11/PSIC-01)

period: from 29/04/2019 to 29/04/2030

2020-present Scientific supervisor in the consortium of the project MAIA

Project: **MAIA**: Multifunctional, adaptive and interactive AI system for Acting in multiple contexts

Funding: Horizon 2020 Framework Programme for Research and Innovation FET-Proactive

Grant: No. 951910

2024-June 2026 Member of GEV 2020-2024

role: Componente del GEV 11b - Scienze psicologiche - VQR 2020-2024

GSD/SSD: 11/PSIC-01; PSIC-01/A - Psicologia generale

Ambito del GEV: GSD 11/PSIC-01, 11/PSIC-02, 11/PSIC-03, 11/PSIC-04

starting: 30 April 2024

2023-present Scientific supervisor in the consortium of the PRIN PNRR 2022

title: A Brain-to-brain coupling: modeling and validating a theory of interactive inference in non-human primates

Grant: No. 20224FESY, CUP B53D23030100001

funding: PRIN PNRR 2022, D.D. 1409 14/09/2022

starting: 30/11/2023

2020-present Scientific Supervisor in the consortium of the THINKAHEAD project

Grant agreement: 820213

Funder: European Research Council (ERC)

Project: Thinking Ahead: human planning from a predictive processing perspective

Total cost: Euro 1'898'757.75

EDITORIAL ACTIVITIES**2023-present Associate Editor**

Frontiers in Neurobotics

ISSN: 1662-5218

www.frontiersin.org/journals/neurobotics#editorial-board

2021-2023 Guest Editor

Applied Sciences

ISSN: 2076-3417

Special Issue in "Deep Learning for Electroencephalography(EEG) Data Analysis".

Section "Computing and Artificial Intelligence".

https://www.mdpi.com/journal/applsci/special_issues/deep_learning_EEG

2020-2021 Guest Editor

Entropy

ISSN 1099-4300

Special Issue in "Probabilistic Inference in Goal-Directed Human and Animal Decision-Making".

Section "Information Theory, Probability and Statistics". https://www.mdpi.com/journal/entropy/special_issues/probabilistic_inference

2021 Guest Editor

Frontiers in Computational Neuroscience

ISSN 1662-5188

Topic: "The Non-Learning Plasticity Hypothesis: Computational models, programmability, re-use and neuromodulation"

2017-2018 Reviewer of Scientific Proposals

European Research Council (ERC) Consolidator Grant 2017 Call

Appointer: European Research Council Executive Agency (ERCEA)

Call: ERC-2017-COG. Panel: PE6 (June 2017 to January 2018).

INSTITUTIONAL RESPONSIBILITIES**2026 Member of Research Contract and Grant Selection Committees**

Institution: Institute of Cognitive Sciences and Technologies - CNR

Calls: IR N. 05/2026 ISTC Roma; CR N. 05/2026 ISTC Roma; CR N. 06/2026 ISTC Roma; CR N. 07/2026 ISTC Roma; BR N. 21/2026 ISTC Roma

Role: Member of the evaluation/examining committees

2026 Responsible for External Collaboration Procedure

Call: Avviso n. 11/ISTC/2026

Institution: Institute of Cognitive Sciences and Technologies - CNR

Activity: Development of computational models, decision-making models and brain-decoding algorithms for identifying and representing neural manifolds across cognitive conditions

2026 Reviewer of Ph.D. Thesis

Institution: University of Rome "Tor Vergata"

Candidate: Nicola Alboré

Program: Ph.D. in Physics - Cycle XXXVIII

Thesis title: Recurrent Dynamics Across Scales: From Neural Computation to Complex Natural Systems

2024-June 2026 Member of GEV 2020-2024

role: Componente del GEV 11b - Scienze psicologiche - VQR 2020-2024

GSD/SSD: 11/PSIC-01; PSIC-01/A - Psicologia generale

Ambito del GEV: GSD 11/PSIC-01, 11/PSIC-02, 11/PSIC-03, 11/PSIC-04

starting: 30 April 2024

2024-present Scientific Advisor for Ph.D. Student

Institution: Sapienza Università di Roma - Dottorato Nazionale in Intelligenza Artificiale

Cycle: XXXIX

Student Name: Lorenzo Ognibeni

Host Institution: ISTC-CNR

Research Program: Analysis and computational modelling of goal-directed tasks with neural and behavioural datasets

2025 Reviewer of Ph.D. Thesis

Institution: Sapienza Università di Roma, Department of Physiology and Pharmacology "Vittorio Erspamer"

Candidate: Stefano Grasso

Program: Ph.D. in Behavioural Neuroscience - Cycle XXXVII

Thesis title: Neuronal Correlates of Motor Control and Economic Decision-Making During Social Interactions in Non-Human Primates

2025 Member of M.Sc. Thesis Defense Committee

Institution: School of Industrial and Information Engineering, Politecnico di Milano

Program: M.Sc. in Mathematical Engineering

Defense Date: 22/07/2025

Candidate: Francesco Maria Mancinelli

Thesis Title: Multi-Agent Decision Making via Active Inference

2025 Member of Ph.D. Defense Committee

Institution: Université Claude Bernard Lyon 1

Defense Date: 10/07/2025

Candidate: Côme Annicchiarico

Doctoral School: ED 476 - NSCo - Neurosciences & Cognition

Cycle: XXXVII - NEUROSCIENCE

Thesis title: Bayesian Modelling of Learning and Failure Mechanisms in Neurofeedback Training

2025 Member of Ph.D. Defense Committee

Institution: Politecnico di Torino

Defense Date: 03/07/2025

Candidate: Leo D'Amato

Doctoral School: Artificial Intelligence

Cycle: XXXVII

Thesis title: Efficient Neural Coding Under Resource Constraints: A Rate-Distortion Theory Perspective

2025 Member of Ph.D. Defense Committee, n 73-2023

Institution: Università di Padova

Defense Date: 17/02/2025

Cycle: XXXVII - NEUROSCIENCE

2025 Responsible for the Call procedure for a TD Researcher position

Call: 331.3 RIC ISTC

Profile: Researcher III Level, Fixed Term

Institution: ISTC-CNR

CUP: B54I19001200006

2025 Responsible for the Call procedure for a TD Researcher position

Call: 331.2 RIC ISTC

Profile: Researcher III Level, Fixed Term

Institution: ISTC-CNR

CUP: B54I19001200006

2024 Reviewer of Ph.D. Thesis

Institution: Ph.D. School in Information Engineering - Università di Padova

Candidate: Giorgia Baron

Cycle: XXXVI

Thesis title: Exploring the large-scale mechanisms orchestrating human brain functioning through dynamic causal modelling of resting-state fMRI

2023-present Scientific Supervisor for Research Fellow

Call: ISTC-AdR-367-2023-RM

Junior Research Fellow: Ph.D. student Lorenzo Ognibeni at ISTC-CNR

Research Program: Free energy principle and the brain: neuronal and phylogenetic mechanisms of Bayesian inference

2022-present Scientific Supervisor for Research Fellow

Call: ISTC-AdR-321-2022-RM

Senior Research Fellow: Dr. Mirco Frosolone at ISTC-CNR

Research Program: Thinking Ahead: human planning from a predictive processing perspective

2023 Responsible for the Call procedure for a TD Researcher position

Call: 400.11.ISTC.PNRR - FAIR project

Profile: Researcher III Level, Fixed Term

Institution: ISTC-CNR

CUP: B53C22003630006

2023 Responsible for the Call procedure for a TD Researcher position

Call: 400.05.RIC.ISTC.PNRR - EBRAINS project Italy

Profile: Researcher III Level, Fixed Term

Institution: ISTC-CNR

CUP: B51E22000150006

2023 Member of the Evaluation Commission for Research Fellow call

Job: Research Fellow

Application: ISTC-361-2023-RM - 11/05/2023

Theme: Probabilistic and neural computational models of sensorimotor functions and experimental validations of models

Structure: Institute of Cognitive Sciences and Technologies - CNR

Project: ThinkAhead - Thinking Ahead: human planning from a predictive processing perspective

2020-present **Member of the consortium - THINKAHEAD project**

Grant agreement: 820213

Funder: European Research Council (ERC)

Project: Thinking Ahead: human planning from a predictive processing perspective

Total cost: Euro 1'898'757.75

2016-2018 **Member of the consortium for the HSFP project**

Funder: Human Science Frontiers Project (HSFP)

grant agreement: RGY0088/2014

Project: Beyond simple choices: computational and neuronal mechanisms for complex spatial behaviors

Team: leaders of the funded project Matthijs van der Meer (University of Waterloo, Canada), Caleb Kemere (Rice University, Houston, USA) and Giovanni Pezzulo (Institute of Cognitive Sciences and Technologies, National Research Council, Rome, Italy)

2019-2023 Member of consortium for the HBP project

Project: Human Brain Project (HBP) - European Future and Emerging Technologies (FET) Flagship project

Funding: second grant agreement (SGA2) and SGA3 - Horizon 2020 Framework Programme for Research and Innovation

Grant number: No. 785907 and 945539

2019-2023 Member of the Consortium for the ALV project

Project: Affordance Landscape Viewer (ALV): Learning new skills by visualization object affordances that only experts can see

Funding scheme: Opportunities in Science and Technology (NICOP)

Funder: Department of the Navy, Office of Naval Research (ONR)

Grant number: N62909-19-1-2017

2022 Member of the Evaluation Committee

Job: Research Fellow

Application: ISTC-ADR-328-2022-RM Ernst & Young - 05/12/2022

Theme: Studio degli effetti del training sportivo offerto dall'associazione sportiva Real Eyes Sport sullo sviluppo sensomotorio di bambini e ragazzi non vedenti ed ipovedenti

Structure: Institute of Cognitive Science and Technologies - CNR

Project: Accordo di collaborazione scientifica con Ernst & Young

2020 Member of the Evaluation Commission for a Research Fellow Position

Job: Research Fellow

Application: ISTC-AdR-282-2020-PD - 04/09/2020

Theme: Probabilistic and neural computational models of sensorimotor functions and experimental validations of models

Structure: Institute of Cognitive Science and Technologies - CNR

Project: Performing Actions in a Changing Environment (PACE) Project - PRIN-2017

Funding: MIUR - Ministero dell'Istruzione, dell'Università e della Ricerca

Timeframe: 01 September 2019 - 31 August 2022

2017-2018 Reviewer of Scientific Proposals

Call Type: European Research Council (ERC) Consolidator Grant 2017 Call

Appointer: European Research Council Executive Agency (ERCEA)

Call: ERC-2017-COG.

Panel: PE6 (June 2017 to January 2018)

SCIENTIFIC MEETINGS AND INVITED CONTRIBUTIONS**2026 Invited Workshop Speaker**

Workshop at Politecnico di Milano.

Contribution title: "On the Structure of Compositional Intelligence: Hierarchical Representations and the Reuse of Neural Computation".

Milan, Italy, 21 January 2026.

2025 Invited Seminar

Dipartimento di Ingegneria Elettrica e delle Tecnologie dell'Informazione, Università degli Studi di Napoli Federico II.

Contribution title: "Computational models for brain representation and decoding".

Naples, Italy, 28 February 2025.

2023 Special Session organizer

ADVANCED IMAGE ANALYSIS FOR BIOMEDICAL APPLICATIONS + MACHINE LEARNING METHODS FOR BIOSIGNAL MODELING AND INTERPRETATION
at IEEE International Conference on
Metrology for eXtended Reality, Artificial Intelligence, and Neural Engineering - IEEE MetroX-RAINE 2023
October 25-27, Milan, Italy

2022 Member of the Scientific Program Committee

IEEE International Conference on
Metrology for eXtended Reality, Artificial Intelligence, and Neural Engineering
IEEE MetroXRAINE 2022
October 26-28, Rome, Italy

Special Session organizer

MACHINE LEARNING METHODS FOR BIOSIGNAL MODELING AND INTERPRETATION
Special Session at the IEEE International Conference on
Metrology for eXtended Reality, Artificial Intelligence, and Neural Engineering
IEEE MetroXRAINE 2022
October 26-28, Rome, Italy
Together with: Prof. Roberto Prevete and Prof. Francesco Isgró

2020 Invited Speaker

International Conference on Robotics and Automation - ICRA 2020 Workshop on Human-Robot Handovers.
Contribution title: "Models for Sensorimotor Communication in Humans and Robot".
Paris 4 June 2020 (virtual conference due to Covid-19 pandemic).

2019 Invited Speaker

Ph.D. School "Italo Gorini" 2019.
Contribution title: "Learning Efficient Codes for Environmental Signal Representation and Control Tasks".
2-6 September 2019. Naples, Italy
<https://www.gmee.org/gorini2019/donnarumma>

2018 Invited Speaker

41st European Conference on Visual Perception (ECVP 2018), in Symposium *Perceiving others, interacting with them: new perspectives in action observation research*.
Contribution title: "Sensorimotor Communication Strategies for Improving Interactive Skills".
26-30 August 2018, Trieste, Italy.
doi:10.1177/0301006618824879

2017 Member of the organization committee

Meeting: *Probabilistic approaches to brain and behaviour*
July 18-23, Praia a Mare (CS), Italy

2016 Member of the organization committee

International Meeting: *Internally generated sequences in the hippocampus*
September 28-30, 2016 - Ariccia (Rome), Italy.

2014 Project meeting

Action, Brain, Language, & Evolution (ABLE) meeting
Workshop for the ABLE project, coordinated by Michael Arbib.
December 1-2, Rome, Italy

Project meeting

Anatomy of Choice meeting
Workshop and Tutorial at Wellcome Trust Centre for Neuroimaging, University College of London, 12 Queen Square
November 6-7, London, Great Britain

2013 **Poster Presentation**

JAM V - 5th Joint Action Meeting, Berlin Humboldt University, Senatssaal, Unter den Linden 6, 10099-Mitte

Contribution title: Sensorimotor Communication: a Theory of Signalling in Online Social Interactions Poster presentation

26-29 2013, Berlin, Germany

Poster Presentation

Symposium on Bayesian Inference: From Spikes to Behaviour at the Bernstein Center for Computational Neuroscience.

Contribution title: Hippocampal forward sweeps and the balance of goal-directed and habitual controllers: a Bayesian approach.

December 9-10, 2011 Tübingen, Germany

2012 Oral Presentation

25th International Conference on Intelligent Robots and Systems (IROS)

Contribution title: Fast incremental clustering and representation of a 3D point cloud sequence with planar regions

October 10th, 2012 - Vilamoura, Algarve - Portugal.

2010 Member of the organization committee

Workshop on Mirror Codes for Social Interactions

June 28-29, 2010 – Capri, Italy.

Invited Presentation

Università degli studi di Milano-Bicocca, Dipartimento di Scienze Umane per la Formazione

Contribution title: *Environmental Maps and neural networks: a simulation study*

In collaboration with the Project Prin 2007, “Il comportamento adattativo dei sistemi biologici e il metodo scientifico: il ruolo dei modelli formali e materiali, le spiegazioni funzionali, gli approcci disciplinari e l'unificazione”

October 6th, 2010 Milan, Italy

2008 Invited joint presentation

Joint presentation with Prof. Guglielmo Tamburrini, Università di Napoli Federico II and Dr. Matteo Santoro, Università di Genova,

Steinberg Auditorium of Carnegie Mellon University, Pittsburgh, U.S.A.

Contribution title: Mutual Knowledge and the Ethics of Human-Robot Interaction

April 3rd, 2008

Oral Presentation

ECAI 2008, 18th European Conference on Artificial Intelligence

Contribution title: CTRNN Parameter Learning using Differential Evolution

July 15, 2008 - Patras, Greece.

2007 Member of the organization committee

Workshop on Medical Imaging

Project: MAGIC-V (Medical Applications on a Grid Infrastructure Connection), INFN Group V, Section of Naples

June 20-22, 2007 - Naples, Italy

Summer School Attendant

SECEViTA - Corso Estivo in Computazione Artificiale e Vita Evolutiva

Summer School in Artificial Life and Evolutionary Approaches

August 31st - September 4th, 2007 - Sampieri (Ragusa), Italy

Oral Presentation

WIVACE 2007, Workshop Italiano di Vita Artificiale e Computazione Evolutiva

Contribution title: A Differential Evolution approach to CTRNN parameter learning

September 6, 2007 - Sampieri (Ragusa), Italy

Invited Discussant

Discussant on the contribution of Prof. Luciano Pietronero, Università di Roma La Sapienza: “La complessità dalle scienze naturali a quelle sociali” at the Quarto Convegno della Associazione Italiana Scienze Cognitive (AISC) *Cognizione, Complessità, Cittadinanza*

November 28-29, 2007 Aula Magna Luiss Guido Carli, Rome, Italy

Poster presentation

ICMP 2007, International Conference on Morphological Computation. Venice, Italy
Contribution title: Virtuality in Neural Dynamical Systems
26-28 March 2007

2006 Member of the organization committee

International Workshop on Ethics of Human Interaction with Robotic, Bionic, and AI Systems:
Concepts and Policies
October 17-18, 2006 - Naples, Italy

TEACHING AND DIDACTIC ACTIVITIES

2024-present Scientific Supervisor

Ph.D. student Lorenzo Ognibeni
Sapienza Università di Roma - Dottorato Nazionale in Intelligenza Artificiale, Cycle XXXIX
Host Institution: ISTC-CNR
Research theme: analysis and computational modelling of goal-directed tasks with neural and behavioural datasets

2021-present Scientific Supervisor

Postdoctoral and research fellows, Ph.D. student
Institution: ISTC-CNR

2019-2022 Scientific Co-supervisor

2 Ph.D. students
Institution: CERN (European Organization for Nuclear Research) - Technology Department

2010-2020 Scientific Co-supervisor

1 Ph.D. student, 3 Master Students
Institution: Università degli Studi di Napoli Federico II

2014 Head of the Course

Analysis and interpretation of brain signals for Brain-Computer Interfaces applications
Fondazione Neurone Onlus, for Cyber Brain project - PON "Ricerca e Competitività" 2007- 2013,
Linea di intervento 4.1.1.4
September 1 - December 31, Napoli, Italy

2011-2012 Teaching support

Contract of 25 hours of Didactic Activities
Course: Laboratory of Programming
Università di Napoli Federico II

2009 Didactic support

Course of *Reti Neurali e Machine Learning Modulo A*, Informatic Degree Course 2009-2010
Università degli studi di Napoli Federico II

Didactic support

Course of *Reti Neurali e Machine Learning Modulo B*, Informatic Degree Course 2009-2010
Università degli studi di Napoli Federico II

2008 Didactic support

Course of *Reti Neurali e Machine Learning Modulo A*, Informatic Degree Course 2008-2009
Università degli studi di Napoli Federico II

Didactic support

Course of *Reti Neurali e Machine Learning Modulo B*, Informatic Degree Course 2008-2009
Università degli studi di Napoli Federico II

2007 Didactic support

Course of *Mente e Macchine*, Informatic Degree Course 2007-2008
Università degli studi di Napoli Federico II

Didactic support

Course of *Reti Neurali*, Informatic Degree Course 2007-2008
Università degli studi di Napoli Federico II

Didactic support

Course of *Elaborazione delle Immagini II*, Informatic Degree Course 2007-2008
Università degli studi di Napoli Federico II

**RESEARCH AND
COLLABORATIONS****2023-present Research Unit Director**

Institution: Institute of Cognitive Sciences and Technologies (ISTC), National Research Council (CNR)
Project: PRIN2020 Project: Free energy principle and the brain: neuronal and phylogenetic mechanisms of Bayesian inference (Grant N.: 2020529PCP)
Units: Department of Neuroscience, Università degli Studi di Padova (Ettore Ambrosini, Antonino Vallesi, Antonino Visalli)
Department of Biomedical Sciences, Università degli Studi di Padova (Aram Megighian, Marco Dal Maschio)
Istituto di neuroscienze (IN), National Research Council (CNR) - (Claudia Lodovichi)
Department of General Psychology, Università degli Studi di Padova - (Maria Elena Miletto Petrazzini)
ISTC-CNR - (Francesco Donnarumma, Giovanni Pezzulo, Francesco Mannella, Antonella Maselli)

2015-present Senior member of the CONAN (Cognition iN ActionN) Lab

Institution: Institute of Cognitive Sciences and Technologies (ISTC), National Research Council (CNR)
Selected projects: "Beyond simple choices: computational and neuronal mechanisms for complex spatial behaviors" Human Frontier Science Program (HFSP) Grant No: RGY0088/2014
Laboratory group and selected collaborations: ISTC-CNR laboratory group including, among others, Giovanni Pezzulo, Domenico Maisto and Ivilin Peev Stoianov
vandermeerlab - Department of Psychological and Brain Sciences at Dartmouth College, Hanover, New Hampshire, USA – Director: Matthijs van der Meer
kemere lab - Rice University, Houston, Texas, USA – Director: Caleb Kemere;
Sapienza Università di Roma - Dipartimento di Fisiologia e Farmacologia "Vittorio Erspamer" (Aldo Genovesio)

**2014-2021 International research group on
Active Inference and models for visual perception**

Units: The Wellcome Trust Centre for Neuroimaging, UCL, London, UK (Karl Friston)
Università G. d'Annunzio, Chieti – Dip. di Neuroscienze e Imaging e scienze cliniche Laboratorio di Neuropsicologia e Neuroscienze Cognitive (Marcello Costantini, Ettore Ambrosini)
ISTC-CNR (Francesco Donnarumma, Giovanni Pezzulo)

**2019-2021 International research collaboration on
Machine Learning Methods For Measurements**

Units: ISTC-CNR (Francesco Donnarumma)
Technology Department, CERN – European Organization for Nuclear Research, 1211 Meyrin, Switzerland (Marco Buzio)
Instrumentation and Measurement Laboratory for Particle Accelerator Laboratory (IMPALab)
Department of Electrical Engineering and Information Technology (DIETI) University of Naples Federico II (Pasquale Arpaia)

2013-2020 **International research group on Joint Action Models**

Units: Università di Palermo - Computer Science Engineering - Haris Dindo
Central European University - Department of Cognitive Science - Günther Knoblich
ISTC-CNR (Francesco Donnarumma, Giovanni Pezzulo, Pierpaolo Iodice)

2011-2016 GOAL (Goal-Oriented Agents) Lab

Institution: Institute of Cognitive Sciences and Technologies (ISTC), National Research Council (CNR) in Rome, Italy

Director: Cristiano Castelfranchi

Main collaborators: Giovanni Pezzulo, Fabian Chersi, Ivilin Peer Stoianov.

Selected projects: European project FP7-ICT-231453 HUMANOBS - Humanoids That Learn Socio-Communicative Skills Through Observation (2011-2012) and the European project FP7-ICT-270108 Goal-Leaders: Goal-directed, Adaptive Builder Robots (2011-2014)

2006-2011 Member of ViNeLab (computer Vision and NEural networks laboratory)

Institution: Department of Physics Università degli Studi di Napoli Federico II, Napoli Italy

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Selected projects: "Actions representations and their impairment" Fondazione San Paolo, Italy, in collaboration with units from Torino, Bari, Genova and Pisa. PRIN2007 - "Modelli computazionali e robotici del comportamento biologico adattativo: problemi metodologici e integrazione multidisciplinare". PRIN2015 "Perception, Performativity and Cognitive Science", and in PON projects like ORCHESTRA (2012-2015), CHIS (2015-2016), SMART HEALTH (2012-2015) e European projects DEXMART (2008-2011) and AIROBOTS (2010-2013)

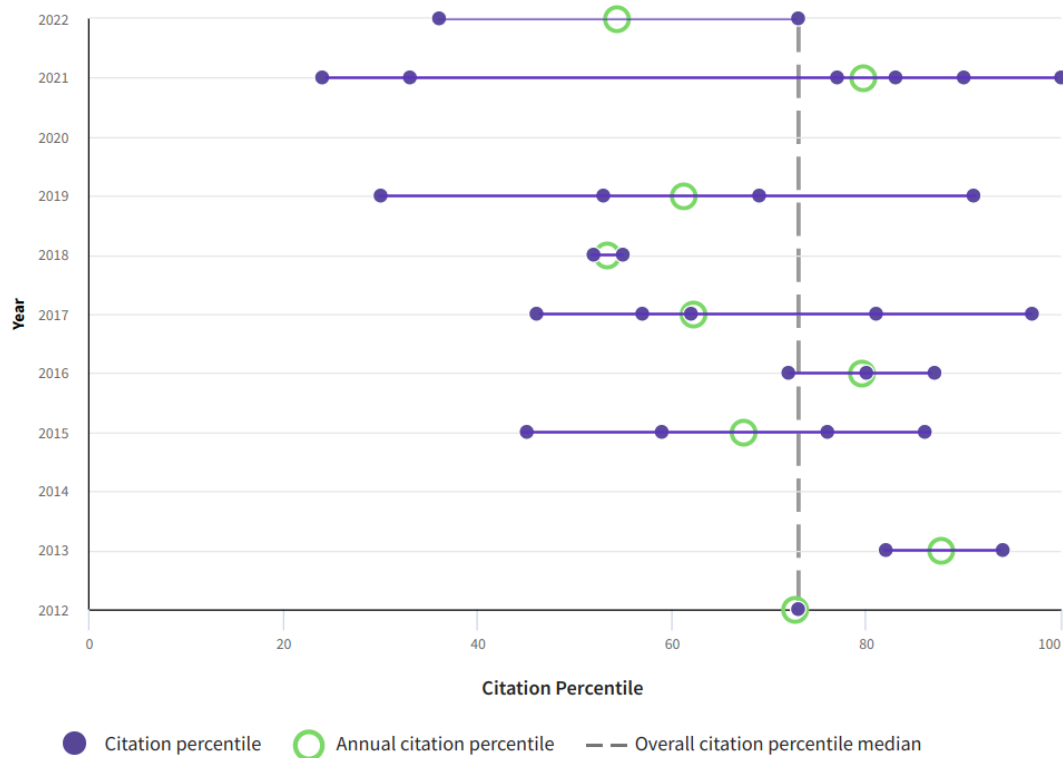


Figure 1: Donnarumma's Citation Percentile in WOS. Citation percentile measures the number of citations of an article against similar papers (see WOS)



Figure 2: Donnarumma's Citation Map. Spread of citations of Donnarumma's papers around the world. (see WOS)

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- [1] Enrico Giralaldi, Mirco Frosolone, Andrew W. Corcoran, Laura Barca, Alessandro Corsini, **Francesco Donnarumma**, Alessandro D'Ausilio, and Giovanni Pezzulo. "False but phonologically plausible linguistic priors induce cross-linguistic auditory illusions and attenuate electrophysiological markers of surprise". In: *Imaging Neuroscience* 4 (2026), IMAG.a.1178. URL: <https://doi.org/10.1162/IMAG.a.1178>.
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